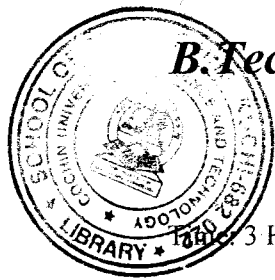


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# **B.Tech. Degree VII Semester Examination November 2018**

## **IT 15-1703 COMPUTER GRAPHICS (2015 Scheme)**



Time: 3 Hours

Maximum Marks: 60

### **PART A** (Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Discuss the applications of computer graphics.
- (b) Consider a line AB with A = (0, 0) and B = (8, 4). Apply a simple DDA algorithm and compute the first four plots on this line.
- (c) Consider a triangle represented by A(0, 0), B(1, 1), C(5, 2). The triangle is rotated by 45° about a point P(-1, -1). Find the coordinates of the new triangle obtained after rotation.
- (d) Consider a unit square centred at origin. The coordinates of the square are translated by a factor (½, 1) and rotated by an angle of 90°. Find the coordinates of the new square.
- (e) Suppose the four rows of array of pixel values for a raster system are 9 3 2, 7 9 8, 2 0 1 and 12 8 2. Find the position of array elements after 90° and 180° counter clockwise rotations.
- (f) Discuss briefly about Bezier curves.
- (g) Discuss briefly about various types of polygon tables using for polygon surfaces representation.
- (h) Translate a cube with vertices (0, 0, 0), (5, 0, 0), (5, 5, 0), (0, 5, 0), (5, 0, 5), (5, 5, 5), (0, 5, 5), (0, 0, 5). Find out the coordinates of the cube after translation. The translation vectors along x, y and z axes are 17, 15 and 18 respectively.
- (i) Explain the method for displaying colour output on a video monitor by using red, green and blue primaries.
- (j) Discuss briefly any two 3D display methods.

### **PART B**

(4 × 10 = 40)

- II. (a) Find out the pixel positions of a line segment with end-points (3, 9) and (10, 14) using Bresenham's line drawing algorithm. (5)
- (b) Demonstrate the midpoint circle algorithm by determining the pixel positions along the first quadrant of a circle with radius 7 cm and center as origin. (5)

**OR**

- III. Apply midpoint ellipse algorithm to find the points of an ellipse with semi major and semi minor axis as 10 and 8 respectively if the centre is at (5, 5). (10)

(P.T.O.)

- IV. (a) A pentagon represented by A(7, 3), B(12, 3), C(12, 10), D(9, 15) and E(7, 10) is translated by  $t_x = 2$  and  $t_y = 4$ . The resultant pentagon is again translated by parameters  $t_x = 9$  and  $t_y = 3$ . Find the composite matrix of the two transformations and also find the new coordinates of the final pentagon. (5)

- (b) Magnify the polygon with vertices A(2, 4), B(3, 2), C(6, 3), D(7, 5) and E(5, 7). Given  $S_x = 2$  and  $S_y = 3$ . (5)

**OR**

- V. Explain Sutherland-Hodgeman polygon clipping with example. (10)

- VI. (a) What are Octrees? How do Octrees differ from Quadtrees? (5)

- (b) Discuss about sweep representation. (5)

**OR**

- VII. (a) Write a short note on constructive solid geometry methods with necessary examples. (5)

- (b) Discuss the different types of fractals. (5)

- VIII. Differentiate between Z Buffer method and A Buffer method in detail. (10)

**OR**

- IX. (a) Illustrate area subdivision method. (6)

- (b) Explain the steps involved in the design of an animation sequence. (4)

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